UNIVERSITÉ DU LUXEMBOURG

Software Engineering 1

Smart Parcel Locker System

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Project Presentation

Last-mile delivery bottleneck

Home deliveries can fail due to the recipient’s absence, leading to costly re-delivery attempts : more pollution and risk of theft or package damage.

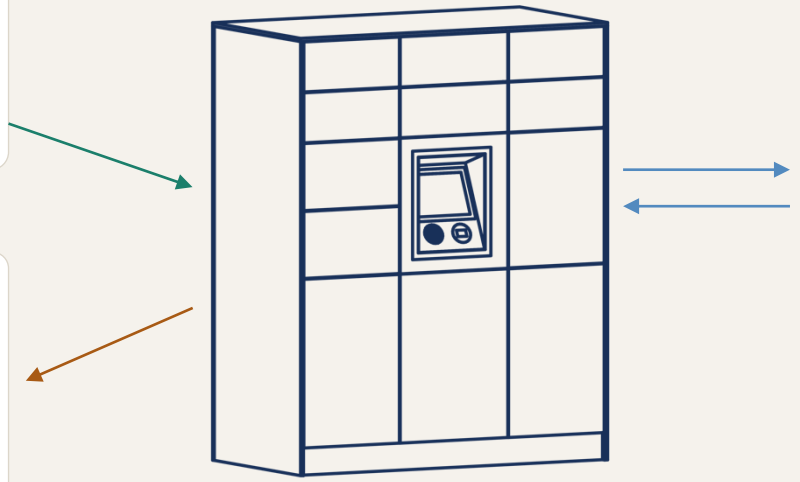
Smart Parcel Locker System : optimized drop-off ; flexible and secure pick-ups ; exceptions handling

Courier

Registers incoming parcels and deposits them in a suitable compartment.

Recipient

Gets a pickup notification and code, then collects the parcel.



Administrator

Monitors availability and status, and reports hardware faults to maintenance.

Requirements

Stakeholders

Primary

Courier, Recipient, Administrator

Secondary

Host businesses, Senders

External

Customs, EU regulations

Hidden

IT team, locker hardware company

Conflict	Tension
Pickup deadline	Long deadlines vs capacity freed quickly
Privacy	Full history for customs vs EU data regulation
Opening hours	24/7 access vs. host business hours
Cost vs quality	Low cost vs SMS alerts and 99.5% uptime

Requirements

Functional requirements (1/2)

Accounts & access

FR_01 Users register with an email address or phone number

FR_02 Guests without an account collect with the code sent by SMS or email

FR_05 Role-based access: recipient, sender, courier, administrator

FR_08 Users can delete their account

FR_13 Couriers sign in at the terminal by app QR code or PIN, tied to their carrier

Parcel lifecycle

FR_03 Standard path: Registered → In Transit → Ready for Pickup → Collected

FR_04 Exceptions: Delayed, Customs Hold, Locker Unavailable, Pickup Expired, Returned to Sender

FR_06 View parcel history, active parcels and receipts

FR_22 When the pickup window ends, the parcel expires and joins the courier's collection list

FR_14 Couriers collect expired, returned and sender-dropped parcels, scanning each removal

Locker network

FR_11 Admins register sites: GPS, address, opening hours, access rules, photos

FR_12 Couriers see locker locations and free compartments of each size

FR_10 Couriers see the parcels assigned to a locker

FR_27 Allocation picks the smallest compartment that fits

FR_25 Automatic alerts: door left open, forced opening, offline terminal, battery power, temperature

Requirements

Functional requirements (2/2)

At the terminal

FR_07 Recipients open with a PIN or tap “open” in the app

FR_26 Senders enter a code and is used a receipt

FR_09 Couriers report exceptions

Notifications

FR_15 Message templates per language

FR_16 A reminder is sent before the pickup deadline

FR_23 Integrates SMS, email and push providers and payment gateways

Payments & pricing

FR_17 Paid extended pickup deadline

FR_19 Card payment for sending and overdue fees

FR_20 Cash on delivery: pickup is blocked until payment is confirmed

FR_21 Configurable pricing per locker, size and region

FR_24 Refunds for failed or cancelled services

Compliance **FR_18** Official customs request export: per parcel or a date range

Requirements

Non-functional requirements

Performance

NFR_05

< 3 s

From a valid pickup code to an open compartment

NFR_04

< 2 s

Page loads and searches, for 95% of requests under normal load

Security

NFR_06

15 min block

Code entry locks after 5 wrong attempts; pickup codes are random and single-use

NFR_01

Secure

Every login endpoint must be secured

Privacy · GDPR

NFR_02

24 h

To remove a deleted account’s data from the system database

NFR_03

Encrypted

Customs-relevant records, in line with GDPR constraints

Requirements

Story mapping

	Deliver (courier)	Collect (recipient)	Manage (admin)
MVP	C0 - C6: authenticate, register, assign or suggest locker, deposit	R0 - R6 : parcel status, pickup code, locker info, open, reminder	A1 - A9: availability, status, mark and restore lockers, history
Release 1	C7 - C9: support, collect expired parcels	R7 - R10: uncollected info, support	A10: expired pickups
Release N		R11 - R12: live tracking, extend deadline, send or return	

UML diagrams

Use case model :

Courier

- Register parcel
- Assign compartment
- Deposit parcel

Recipient

- Track parcel
- Receive pickup notif & code
- Pickup parcel

Administrator

- Visualise availabilities
- Receive operational status
- Change locker status
- Report to maintenance

Shared

Authenticate · Report issue

Secondary actors

Locker hardware · Notification gateway · Maintenance team

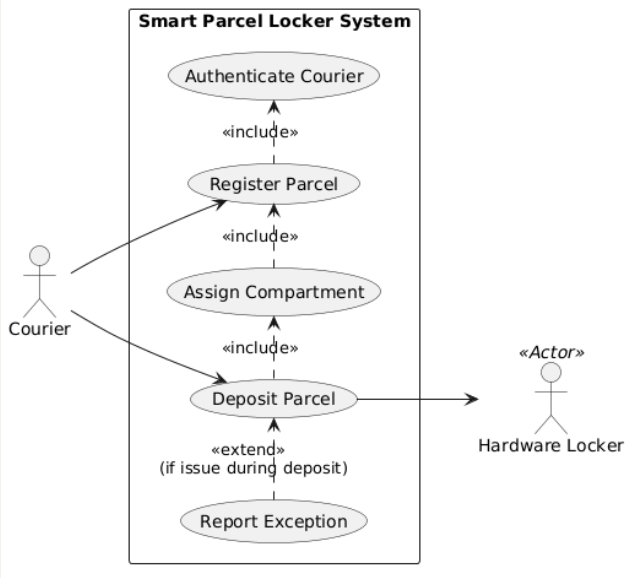
Cross-actor link

Deposit triggers the pickup code the recipient receives

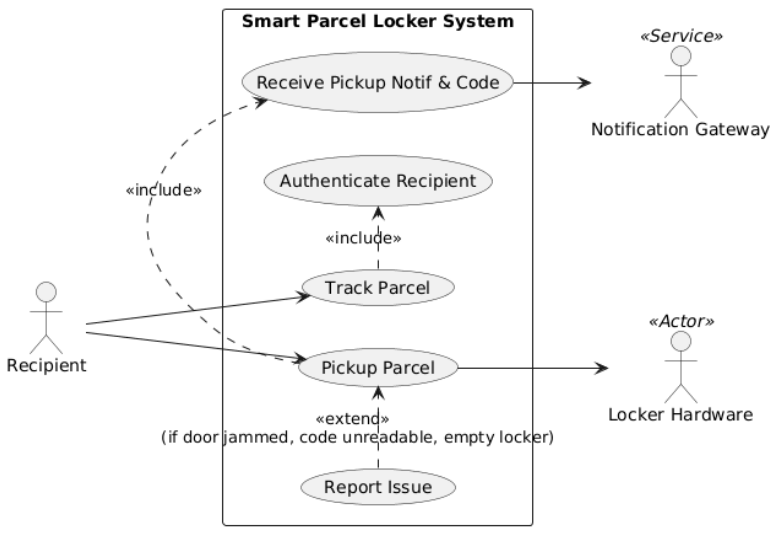
UML diagrams

Use cases diagrams :

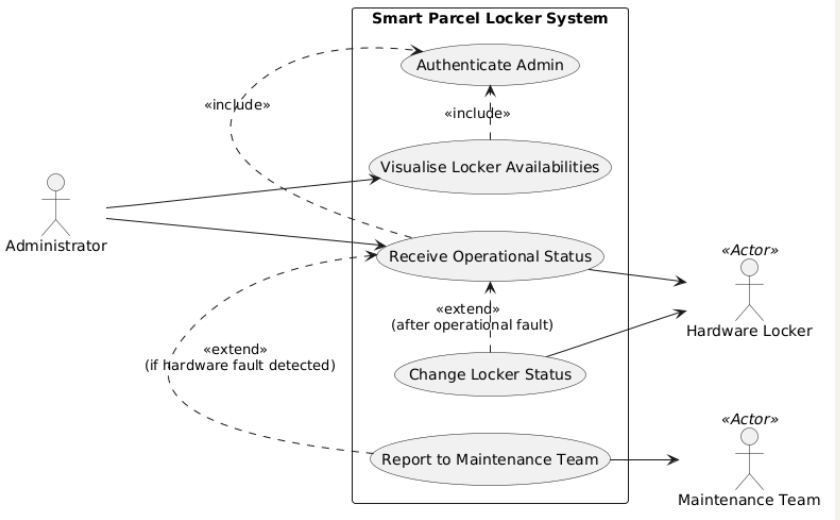
Courier



Recipient



Administrator



Use case description

Example · Deposit a parcel

PRECONDITIONS Authenticated courier • Registered parcel • Assigned locker • Open location

MAIN SUCCESS SCENARIO

1. Find assigned locker
2. Scan or enter parcel code
3. Assigned compartment opens
4. Place parcel & close door
5. System confirms deposit

ALTERNATE FLOW · DOOR FAILURE

Fault detected or reported → locker marked unavailable → another suitable free locker is assigned

POSTCONDITION · SUCCESS

System updates parcel status, recipient notified with a pickup code

UML diagrams

Global use case diagram

How independent workflows connect and cooperate

01 CROSS-ACTOR HANDOFF

Courier deposit ➡ **Gateway sends code** ➡ **Recipient collection**

One completed action enables the next actor's workflow.

02 SHARED OPERATIONS

Authenticate · Report Issue · Reusable across roles

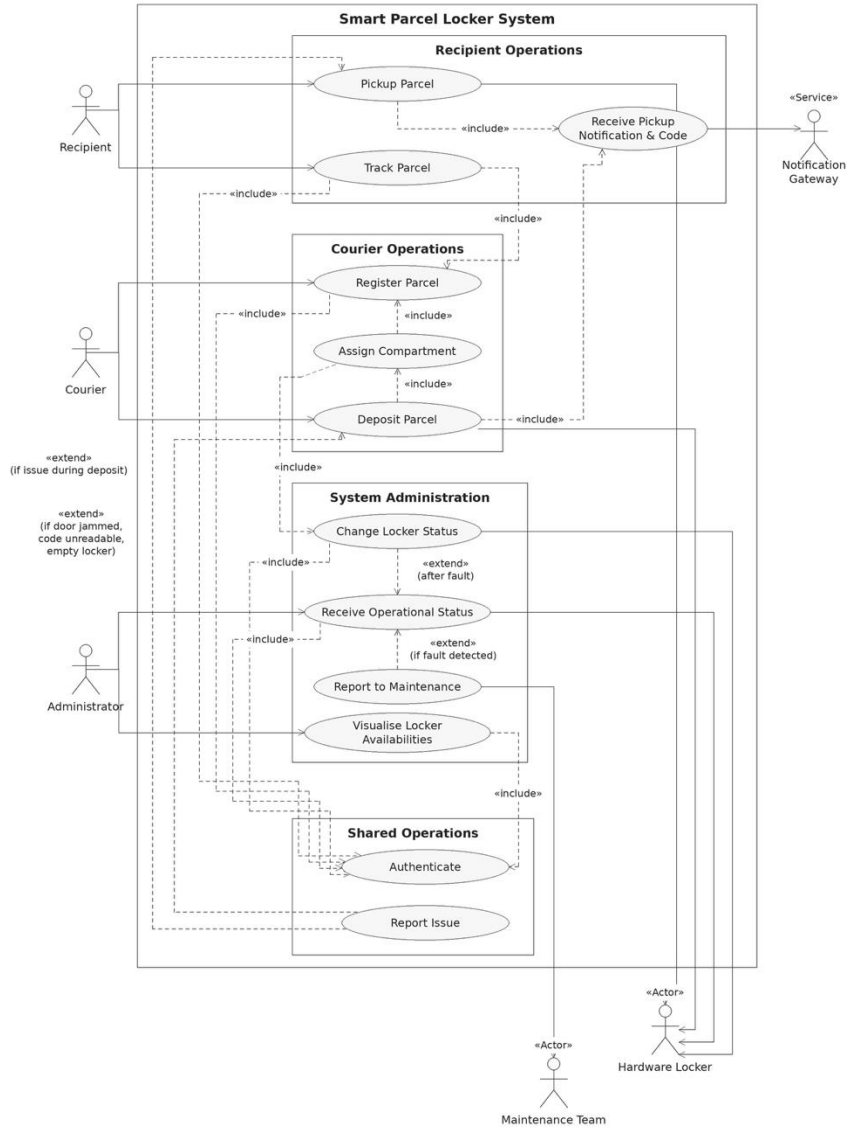
03 LOCKER STATUS & FAULT ESCALATION

Hardware status → Administrator → Maintenance

Physical faults feed monitoring and response.

SYSTEM BOUNDARY

Use cases inside · hardware, gateway & teams outside



UML diagrams

Class diagram

People and roles

Abstract **User**, specialised by **Courier**, **Recipient** and **Administrator**. Couriers and administrators sign in with a **DeliveryCompany** account.

Parcels and their records

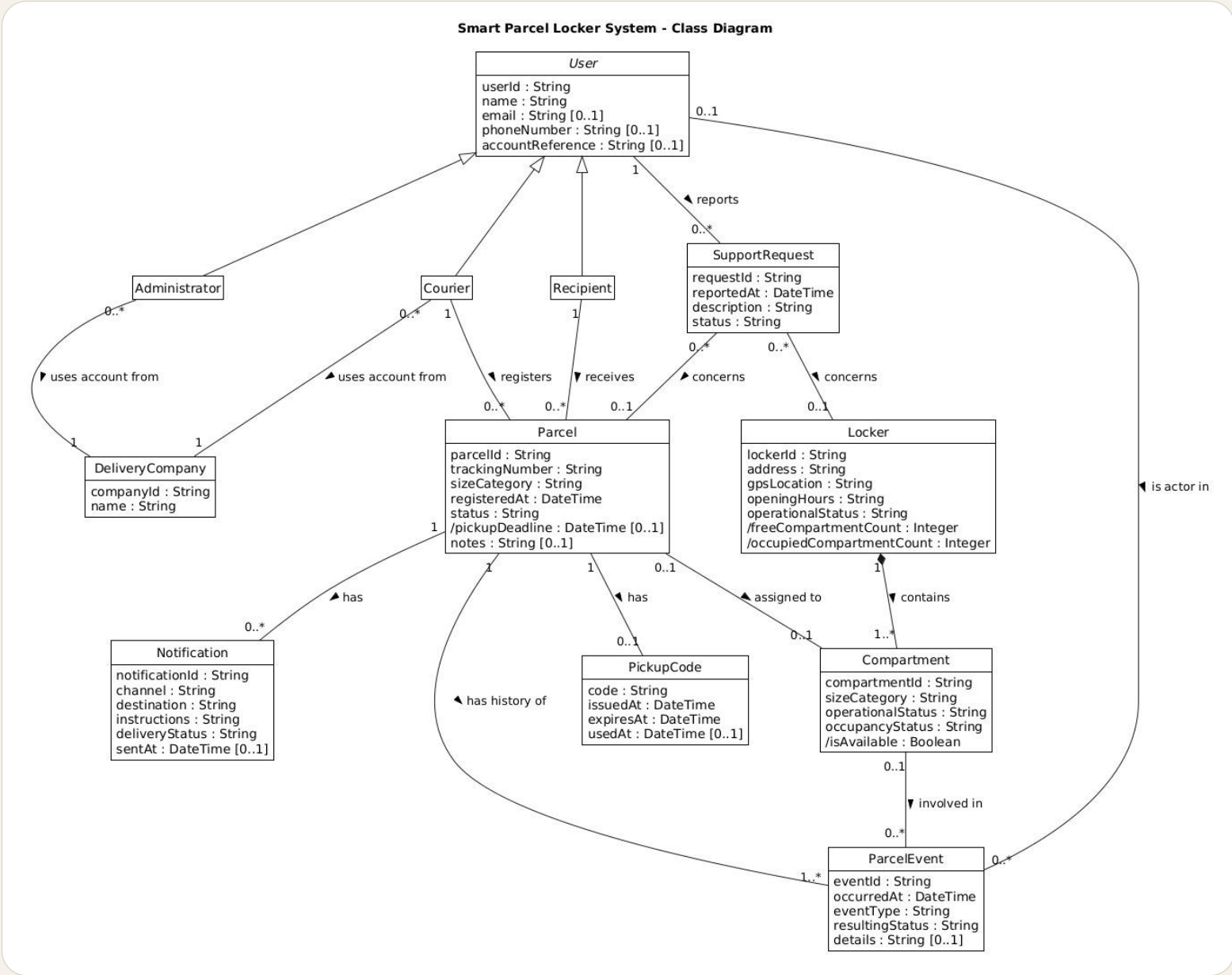
A **Parcel** has 0..1 **PickupCode**, 0..* **Notifications** and 1..* **ParcelEvents** (its history) and is assigned to 0..1 **Compartment**.

Lockers

A **Locker** is composed of 1..* **Compartments**; its free and occupied counts are derived, not stored.

Support

Any User can report a **SupportRequest**, which concerns 0..1 Parcel and 0..1 Locker.



UML diagrams

Behavior modelling :

User | Boundary | Controller | Service | Repository

Courier

Register and assign

Recipient

Receive pickup information and code

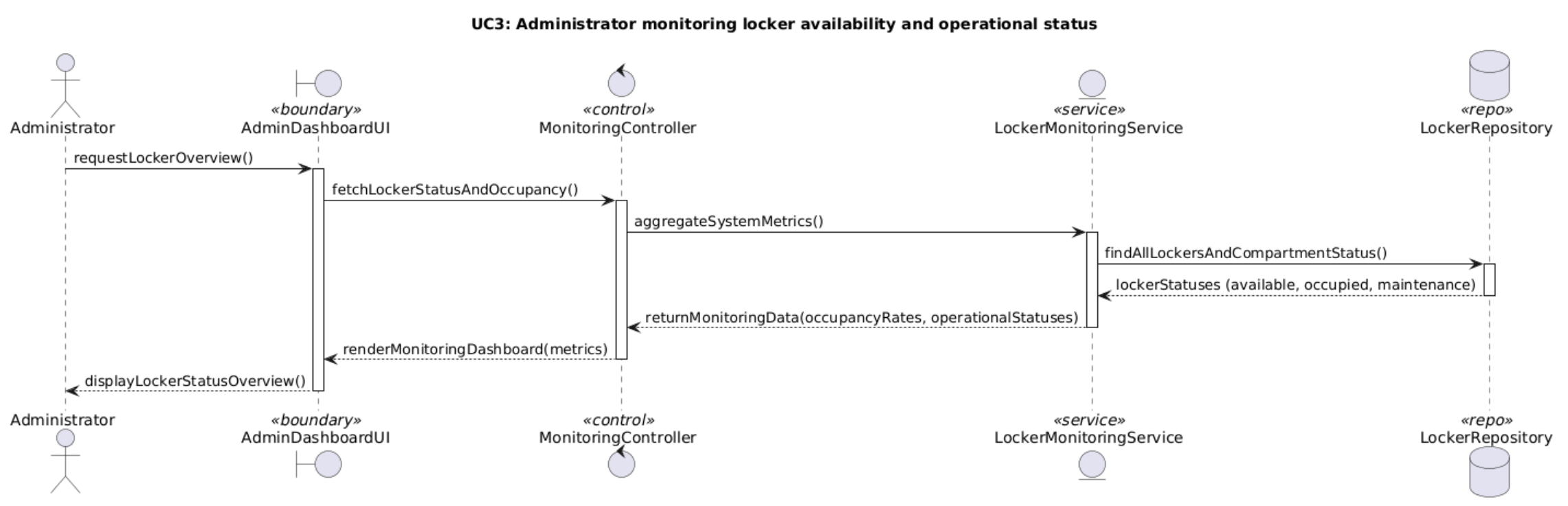
Administrator

View availability and status of lockers

UML diagrams

Sequence diagrams :

Example: administrator monitors lockers



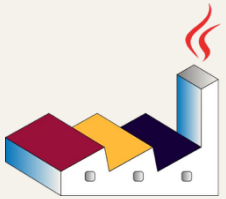
Project management

Methodology

Tools









AI*

Sprints

During practical sessions :

- Review of completed work
- Identification of unfinished tasks and assign them
- Planning the next iteration : define goals and tasks to do

Process

- Shared documents + updates, backlog and work breakdown in the group chat
- Peer reviewing during sprint
- Adjust based on the reviews : work division, planning, functionalities

Project management

Contribution

Deliverables	Members
User stories	Andrès : writing, adding on GitHub , prioritization Serhii : writing Vanathy : prioritization
Requirement analysis (stakeholders, (non) functional, activities)	Andrès, Serhii
Story mapping	Vanathy : Courier and recipient Andrès and Serhii : Admin
Use Case Modelling	Vanathy : initial drawings, textual descriptions, final diagrams Christos : Courier and global system diagrams Serhii : Admin’s diagram and textual description Raed : Recipient’s diagram
Domain Modelling	Andrès
Behavior Modelling	Vasileios : initial drafts Vanathy : final diagrams

Next steps

Refine the models

Apply feedback, align user story IDs, add missing elements

Improve the framework

Use a task board to have a better visualization of our progress

MVP

Implementation…

Thank you for your attention !

Use of AI

- Plant UML code for the diagrams (Gemini)
- Identify the “steps” between Entity, Control and Boundary in the sequence diagrams (Gemini)
- Generate the template of the presentation support (Claude)
- Generate the template of the presentation slide (ChatGPT)